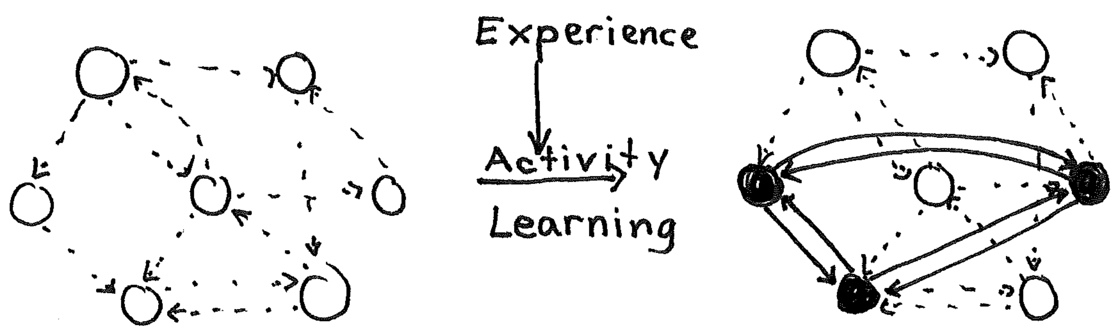
Explicit memory:

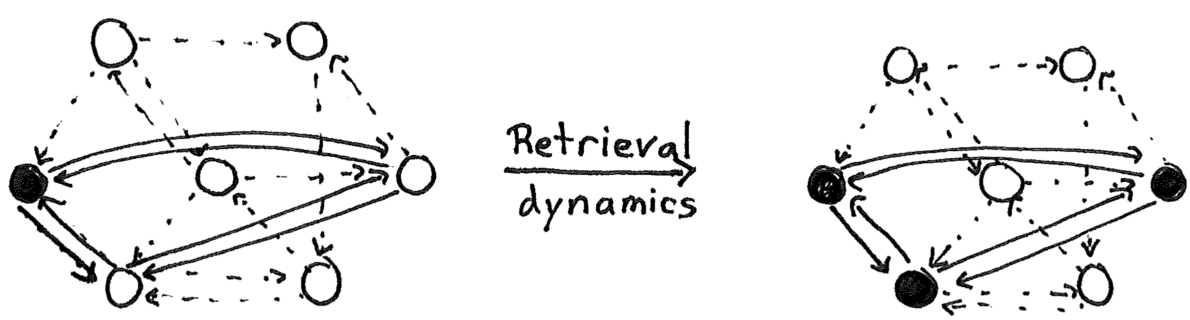
Everyone can think back to recent lectures and recall some things that we covered. How could the brain recall this information?

Hebbian assemblies:

During memory encoding, connections are formed between active neurons.



Recall could subsequently occur by activating a subset of these previously active neurons.



Since retrieval dynamics recreate the same pattern of neural activity present during encoding, you can relive the experience, extract content from it, and so on. In rodents, reactivating hippocampal neurons active during fear conditioning leads to freezing behavior (memory engram).

Hopfield model:

Hopfield built an explicit neural network with these properties. It was based on the physics of magnetic systems, so its formulation is a bit different than what we're used to. Physics Nobel in 2024!

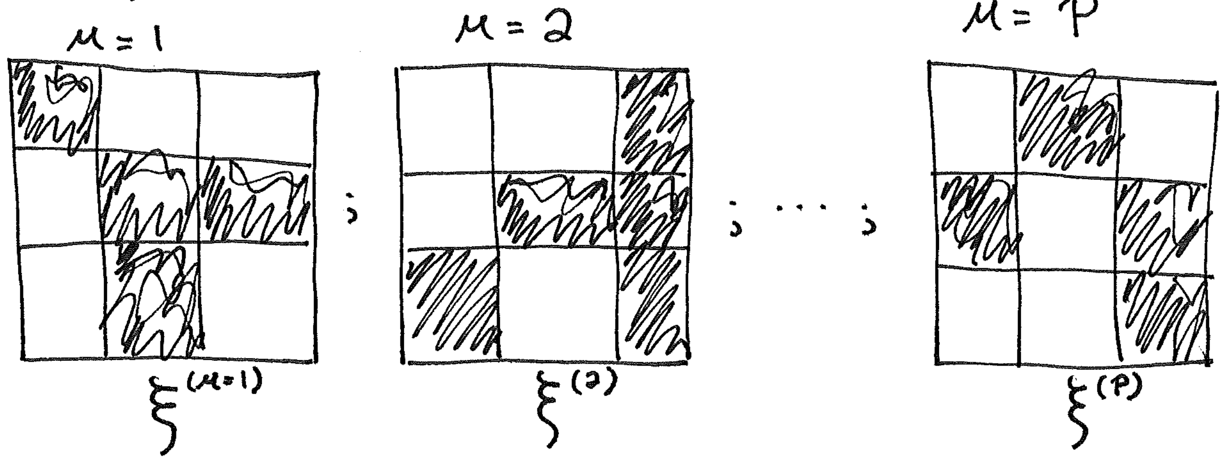
Network dynamics:

- Choose a random neuron, i, for update
- Update state of neuron by

$$\underbrace{S_i}_{\text{Binary state of neuron}}(t+1) = \underbrace{\mathrm{sgn}}_{\pm 1}\Big(\textstyle\sum_{j=1}^{N} \underbrace{W_{ij}}_{\text{Synaptic weights}} S_j(t)\Big)$$

- Iterate until convergence.

Goal: Store P patterns as attractors of network dynamics. (NOTE: This depicts 9 neurons in 2D).

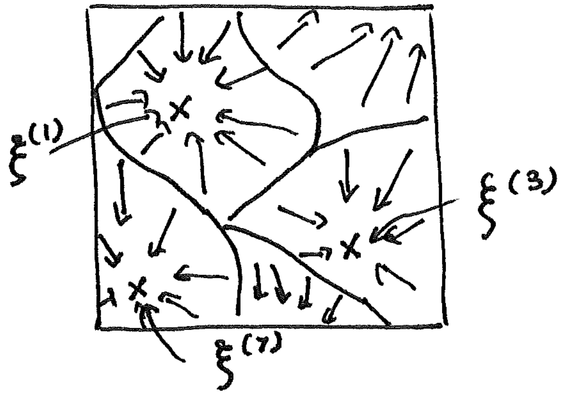


$$P(\xi_i^{(\mu)} = 1) = P(\xi_i^{(\mu)} = -1) = 1/2 \Rightarrow \langle \xi_i^{(\mu)} \rangle = 0$$

$$P(\xi^{(1)}, \xi^{(2)}, \dots, \xi^{(P)}) = \prod_{\mu=1}^{P} \prod_{i=1}^{N} P(\xi_i^{(\mu)})$$

"Ising Model"

$$E = -\frac{1}{2}\sum_i \sum_{j \neq i} \underbrace{W_{ij} S_i S_j}_{\text{Dynamics minimize energy function}} ;$$



$\rightarrow$: Dynamics from model.

$\times$: Attractor state, $\xi^{(\mu)}$

$-$: Basins of attraction

Strategy: Set synaptic weights symmetrically by Hebbian scheme.

$$W_{ij} = \frac{1}{N}\sum_{\mu=1}^{P} \underbrace{\xi_i^{(\mu)} \xi_j^{(\mu)}}_{\substack{\text{"1" if same} \\ \text{"-1" if different}}} , \quad i \neq j$$

$$W_{ii} = 0$$

P = 1 example:

$$S_i(t+1) = \operatorname{sgn}\left(\sum_{j=1}^{N} W_{ij} S_j(t)\right)$$

$$= \operatorname{sgn}\left(\frac{1}{N}\sum_{j\neq i}^{N} \xi_i^{(\mu)} \xi_j^{(\mu)} S_j(t)\right) \qquad \left[u \mathbin{:_i} v \equiv \sum_{j\neq i} u_j v_j\right]$$

$$= \operatorname{sgn}\left(\xi_i^{(\mu)} \left(\xi^{(\mu)} \mathbin{:_i} S(t)\right)\right)$$

$$= \underbrace{\operatorname{sgn}\left(\xi_i^{(\mu)}\right)}_{\xi_i^{(\mu)}} \underbrace{\operatorname{sgn}\left(\xi^{(\mu)} \mathbin{:_i} S(t)\right)}$$

If +1, $S_i(t+1) = \operatorname{sgn}(\xi_i^{(\mu)})$, and dot products gets bigger. Memory retrieval!

If −1, $S_i(t+1) = -\operatorname{sgn}(\xi_i^{(\mu)}) = -\xi_i^{(\mu)}$, and dot product gets smaller. "Spurious attractor" at $-\xi^{(\mu)}$!

Starting in $\pm\xi^{(\mu)}$, the system stays there.

In general, the original Hopfield model has spurious attractors at odd mixture states

1-mixture: $\mathrm{sgn}(\pm \xi^{(\mu)})$

3-mixture: $\mathrm{sgn}(\pm \xi^{(\mu_1)} \pm \xi^{(\mu_2)} \pm \xi^{(\mu_3)})$

$\vdots$

Memory capacity: How many random patterns can be stored as attractors? Suppose $S(t) = \xi^{(\mu)}$. Does it stay there?

$$S_i(t+1) = \mathrm{sgn}\left(\sum_{j=1}^{N} W_{ij}\, \xi_j^{(\mu)}\right) = \mathrm{sgn}\left(\frac{1}{N}\sum_{j\neq i}^{N} \sum_{\nu=1}^{P} \xi_i^{(\nu)} \xi_j^{(\nu)} \xi_j^{(\mu)}\right)$$

$$= \mathrm{sgn}\Bigg(\underbrace{\frac{N-1}{N}\, \xi_i^{(\mu)}}_{\text{Signal from } \mu} + \underbrace{\frac{1}{N} \sum_{j\neq i}^{N} \sum_{\nu \neq \mu}^{P} \xi_i^{(\nu)} \xi_j^{(\nu)} \xi_j^{(\mu)}}_{\text{Interference from } \nu \neq \mu.}\Bigg)$$

$$= \mathrm{sgn}\left(\frac{N-1}{N}\, \xi_i^{(\mu)} + \frac{1}{N}\, \eta_i^{(\mu)}\right)$$

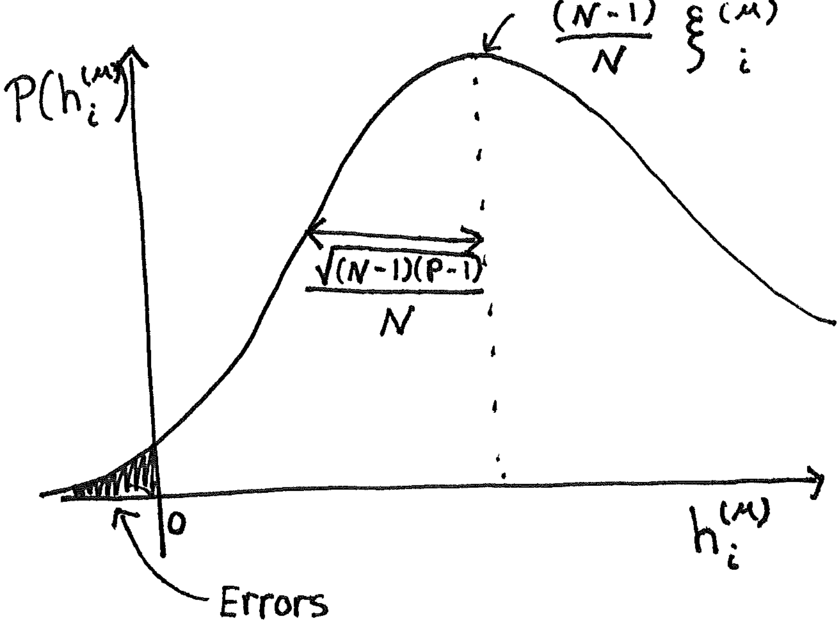
$$\langle \eta_i^{(\mu)} \rangle = 0$$

$$\langle (\eta_i^{(\mu)})^2 \rangle = \Big\langle \sum_{j\neq i} \sum_{\nu\neq\mu} \sum_{k\neq i} \sum_{\rho\neq\mu} \xi_i^{(\nu)} \xi_j^{(\nu)} \xi_j^{(\mu)} \xi_k^{(\rho)} \xi_i^{(\rho)} \xi_k^{(\mu)} \Big\rangle$$

$$= \sum_{j\neq i, k\neq i, \nu\neq\mu, \rho\neq\mu} \underbrace{\langle \xi_i^{(\nu)} \xi_i^{(\rho)} \rangle}_{\delta_{\nu\rho}} \underbrace{\langle \xi_j^{(\mu)} \xi_k^{(\mu)} \rangle}_{\delta_{jk}}{}' \underbrace{\langle \xi_j^{(\nu)} \xi_k^{(\rho)} \rangle}_{1 \text{ (b.c. of other } \delta \text{ terms)}}$$

$$= (N-1)(P-1)$$

Define $h_i^{(\mu)} = \sum_{j=1}^{N} W_{ij}\, \xi_j^{(\mu)}$. The picture is



As $N \to \infty$, mean $\to \xi_i^{(\mu)}$, $\sigma \to \sqrt{\frac{P-1}{N}}$. Substantial errors start to occur when

$$P = \alpha N,$$

so the memory capacity is proportional to the number of neurons.

Amit, Gutfreund, and Sompolinsky leveraged their knowledge of disorded magnetic systems to rigorously derive the proportionality constant as

$$\alpha_c = 0.138$$

Below $P_c = \alpha_c N$, there were attractor states differing from the desired patterns by at most 1.5%. Above P_c, the attractor states were uncorrelated to the patterns. This abrupt change is due to a "spin glass" phase transition.

Other associative memory models:

Hopfield opened the floodgates, and many physicists started to study neural network models of memory. These models sometimes had better properties than Hopfield's. My favorite is due to Buhmann, Divko, and Schulten. It is

- $S_i(t) \in \{0, 1\}$, (active vs. inactive)
- $P(\xi_i^{(\mu)} = 1) = a$, (sparse patterns)
- $W_{ij} = \underbrace{\frac{1}{a(1-a)N} \sum_{\mu=1}^{p} (\xi_i^{(\mu)} - a)(\xi_j^{(\mu)} - a)}_{\text{Hebbian}} - \underbrace{\frac{\gamma}{aN}}_{\text{Global inhibition}}$
- $P(S_i(t+1) = 1) = \frac{1}{1 + e^{-(h_i - u)/T}}$ (stochastic update)

Its nice properties include a parameter regime where

- Attractor at $S(t) = 0$.
- Only other attractors at $\xi^{(\mu)}$.
- Memory capacity and information storage are near optimal for small a.